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Examiners' Report Summer 2026

Pearson Edexcel GCE

In Mathematics (9MA0)

Paper 31 Statistics

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Summer 2026
Publication Code 9MA0_31_2606_ER

General Comments

This proved to be an accessible paper with nearly all learners able to start each question. The first four questions provided many opportunities for all learners to pick up some marks but questions 5 and 6 provided a little more challenge and provided some good discrimination.

Comments on individual questions

Question 1

Part (a)

This was a fairly standard starter requiring learners to identify a pair of mutually exclusive events from the given Venn diagram. It was usually answered well but some answers still confused "events" with "the probability of an event" and wrote down P(*A*) and P(C).

Part (b)

This part was answered successfully by most learners and most found the correct value for *p*.

Part (c)

The usual approach here was to form an equation for P(*B*) using the independence condition. The simplest equation $0.3 \times \mathrm{P}(B) = 0.18$ was often seen and quickly led to the correct answer. Some responses used $0.3 \times (0.18 + \mathrm{r} + 0.15) = 0.18$ and solved to find *r* but then did not follow through with what the question asked, i.e., to find P(*B*), and the accuracy mark was not awarded. Where difficulties arose it was usually over combining the information that *A* and *B* were independent with the fact that *C* was a subset of *B*. Some thought $\mathrm{P(A)} = 0.12$ or $\mathrm{P(B)} = r + 0.18$ or $r + 0.15$

Part (d)

This was usually answered very well with most responses identifying the required probability with *q*. However there were a number of incorrect responses based on the calculation $1 - 0.6 - 0.3 = 0.1$

Part (e)

Although there were many fully correct answers here this part did prove more challenging for some.

In (i) some learners could simply write down the correct answer showing a good understanding of the concept. Some learners encountered difficulty when applying the conditional probability formula whilst some, who had a correct formula, assumed that *B* and *C* were independent and simply cancelled out their P(C) probabilities. In part (ii) many learners did achieve a method mark for giving $\frac{P(B \cap C)}{P(B)}$ and those who realised that the numerator was simply P(C) were usually able to complete this part successfully.

Question 2

Part (a)

Learners were required to name a sampling technique from a contextual description. They were usually successful in identifying one of the two acceptable answers, with "opportunity" sampling more commonly seen than "convenience" sampling. Some realised that a non-random technique was being used and a common incorrect answer was "quota sampling". Whilst slight spelling errors were condoned we did expect the correct terminology as given in the specification to be used and terms such as "opportunistic" or "convenient" were not allowed.

Part (b)

There were fewer correct responses seen here where learners were being assessed on assumptions required for a binomial distribution to be a suitable model. Acceptable answers could focus on independence or the need for constant probability but required the use of precise language and contextual detail. Some learners stated that “faults are not fixed” instead of “the probability of a fault is not constant” and others wrote “the probability of faulty items are independent” instead of “faulty items do not occur independently” or “the probability of a faulty item occuring is constant”. Others simply thought that the sample size was too small or mentioned that the sampling was not random without applying this to the context.

Part (c)

Most learners made a start to this fairly routine binomial hypothesis test. The hypotheses were usually stated correctly in terms of *p* though some used *m* and occasionally we saw a two-tailed test or a test using the lower tail. The correct model was usually stated but, where it was omitted, we could often infer it from subsequent working. It is good practice to state the model being used and label the probability. Many found $P(X \geq 3)$ but there were a number of incorrect probabilities seen such as $P(X = 3), P(X > 3)$ or $P(X \leq 3)$. A minority of learners attempted to use a critical region approach, though this was not intended as the tables do not accommodate it, and sometimes there were no supporting probabilities provided. Giving a correct conclusion to a hypothesis test is still an area of confusion for some learners. Some rejected the null hypothesis despite having a probability greater than the significance level. Others realised that there was insufficient evidence to reject the null hypothesis but did not explain clearly and precisely, in context, what this means.

Question 3

Part (a)

This large data set question was attempted by nearly all learners and part (i) was usually answered correctly. There were only a few responses giving incorrect units such as ml, cm^3 or dm. In contrast, part (ii) was rarely answered correctly with many simply leaving it blank or giving the commonly seen response of “tr”. Nevertheless, a number confidently stated the correct answer thus clearly demonstrating some familiarity with the large data set.

Part (b)

This standard calculation of mean and standard deviation from summary statistics was answered very well by most learners with clear working and appropriate use of the relevant formulae. A few errors were seen in (ii) typically from use of an incorrect formula for standard deviation, failing to square the mean or forgetting to square root the variance.

Part (c)

Here learners were required to interpret a given p-value and draw an appropriate conclusion. There were many misconceptions seen. Some learners thought the p-value was the significance level or treated it as a percentage. Others thought the question was about correlation and compared r with zero and a few gave lengthy descriptions of weather conditions in Jacksonville between 1987 and 2015 or discussed unrelated climate factors such as hurricanes or global warming, ignoring the p-value entirely. However, there were a good number of learners who compared the p-value with a sensible significance level and often went on to give a correct conclusion related to Chris’ belief. Some realised that there was evidence of a change in Daily Total Rainfall but did not specify a direction simply writing “changed” instead of “increased”

Question 4

Part (a)

Most learners gave a correct description of the correlation, often describing it as weak or strong positive correlation but the essential feature that it was “positive” was usually included. A few simply described the relationship between the two variables without identifying the nature of the correlation.

Part (b)

Interpreting the gradient of the regression line was a little more challenging. Some simply identified this as the average velocity or speed of the object whilst others explained that the displacement increased by 39.6 (cm) each second or gave an equivalent definition in terms of rate of change. Occasionally the gradient was incorrectly interpreted as acceleration or an interpretation of the correlation, rather than the gradient, was given.

Part (c)

Most were able to use the regression line to estimate the displacement at 4 seconds, either by reading off from the graph or substituting into the equation. Many stated that the estimate was likely to be an overestimate but their reason lacked enough detail or precision. A good number of learners explained that, because the data points close to $x = 4$ lie below the regression line, therefore the estimate was likely to be an overestimate. Others simply stated that there were more points below the line than above it which did not provide sufficient justification because the points around $x = 4$ could be above the line in such a scenario.

Part (d)

Many learners realised that the points seem to follow a curve rather than a straight line and often suggested that a quadratic or exponential curve might provide a better model. Some simply echoed Nat’s belief and said that a non-linear model was required without giving a suitable reason. A few thought that the reason was that the correlation coefficient was only 0.97 and not 1.

Part (e)

There were many good suggestions here for suitable expressions in Nat's refined model with x^2 and e^x being the most common. Some gave expressions including an unknown constant such as e^{kx} or a^x, however Nat could not use these, with the data, to form the new model and they were not accepted. A few suggested $\log x$ or $\sqrt{x}$ but these had the wrong curvature for these data.

Question 5

Part (a)

Many were able to recall some of the assumptions required for a binomial distribution but selecting a suitable reason here proved challenging. Some were distracted by the coin being biased and thought this meant that the probability of a head would not be constant. Others thought that the spins (trials) were not independent but a good number honed in on the fact that there wasn't a fixed number of trials.

Part (b)

Most learners were able to list the required cases (TTHH, THTH and HTTH) and then work out the probability for each case and show that this led to the printed answer. Some saw that $\frac{2}{5}\times\left(\frac{3}{5}\right)^3=\frac{54}{625}$ and then multiplied by 2 to reach the printed answer which was not awarded any marks. Many realised that the number of heads in the first 3 spins had a binomial distribution and used $\binom{3}{1}\frac{2}{5}\times\left(\frac{3}{5}\right)^2\times\frac{2}{5}$ which was acceptable.

Part (c)

Attempts at finding the probability distribution for X were encouraging. Most learners realised that the sample space was simply 2, 3, 4 and 5 and usually they could find $P(X = 2)$ and $P(X = 3)$ quite easily. Many learners used less efficient approaches to find $P(X = 5)$. Instead of simply using $1 - P(X = 2 \text{ or } 3 \text{ or } 4)$ they chose to try and work out the probabilities of all 10 cases, which invariably led to errors.

Part (d)

The introduction of the new random variable H proved to be challenging for some learners but many realised that they needed 4 tails and one head and were able to give a probability of the form $\left(\frac{3}{5}\right)^4 \times \frac{2}{5}$, however some forgot that there was more than 1 case of 4 tails and one head and so could not make further progress. Some realised that they could use a binomial distribution again and found the correct answer using that approach.

Part (e)

Combining parts (b) and (d) in a conditional probability proved challenging. Those learners who thought through the meaning of the expression often realised that it was impossible and simply wrote down the answer 0 but some embarked on lengthy calculations often leading to a non-zero answer.

Part (f)

Those learners who had interpreted part (e) correctly often went on to identify that the two events were mutually exclusive although a few thought it meant they were independent. Some learners who did not answer part (e) correctly were still able to give a correct response here.

Question 6

Part (a)

Most learners demonstrated some familiarity with the normal distribution though part (a) proved quite challenging. Many did not draw a diagram and often after standardising with 530 or 550 , 540 and s, went on to use a *z* value of 0.8416 (the 20% tail value). A clear diagram should have helped them avoid this error. Those who realised that the probability in each tail was 0.1 usually chose a suitable *z* value though often they used 1.28 or 1.282 rather than the full value from tables or their calculator. By not working with an accurate enough value for *z* often led to an answer that was not correct to 3 decimal places, some with an accurate value for *z* ignored the instruction to use 3 decimal places so were not awarded the accuracy mark.

Part (b)

Some learners realised that $P(C < 530) = 0.1$ and were able to use the binomial model $\mathrm{B}(250, 0.1)$ to write down the mean. Others used the normal distribution with their standard deviation found in (a) to work out the probability and then were able to multiply by 250 to find the estimate and we allowed an answer which rounded to 25 to score both marks.

Part (c)

Most learners demonstrated that they could use their calculators to find this simple probability from a normal distribution. There were a few learners who did not attempt this question and a small minority who did not give their answer to at least 3 significant figures as instructed on the front of the question paper.

Part (d)

It was clear that many learners knew about using a normal distribution to carry out a hypothesis test. The hypotheses were usually correct with only a rare minority failing to label the alternative hypothesis correctly or using some letter instead of m. Most stated the correct model, $\bar{Y} \sim N\left(275, \left(\frac{11}{6}\right)^2\right)$ though a few used the standard deviation instead of variance and a small minority had the mean as 271.7. A variety of different approaches were used here: most found $P(\bar{Y} < 271.7) = 0.0359 \ldots$, some found a critical region $\bar{Y} \leq 271.4$ and a few used $Z = -1.8$ There were a number of responses which were not awarded the final mark for an incorrect comparison such as $0.036 < 0.05$ and a few who thought that because $0.036 > 0.025$ that meant they should reject H_0. However, there were plenty of learners who demonstrated a good knowledge of this topic and gave a correct conclusion in context mentioning Fen's belief or the volume of yoghurt.